

Algebra 1

Unit 13

Lesson 1 Homework

Name_____

Date_____

Period_____

The Singh family is building a pool. They were approved by their bank for a loan of \$85,000 with a simple interest rate of 2.25%.

1. Write an equation that models the Singh's loan.

2. Rewrite the equation in slope-intercept form.

3. Complete the table.

Key Feature	Value	Interpretation in Context
y-intercept		
slope		

4. What is the total the Singh's will have to pay if they pay off the loan in 5 years?

5. How many years did it take the Singh's to pay off the loan if the total they paid was \$109,000?

Sophie just graduated college and is starting her first job. She is investing money into a savings account to plan for retirement. She has \$10,000 to invest in a savings account with a 4.50% APR.

6. Write an equation that models Sophie's savings account.

7. Write the equation in slope-intercept form.

8. Complete the table.

Key Feature	Value	Interpretation in Context
y-intercept		
slope		

9. How much money will be in the savings account if Sophie wants to retire 30 years later?

10. How many years would Sophie have to work to retire with \$30,000?

Algebra 1

Unit 13

Lesson 2 Homework

Name _____

Date _____

Period _____

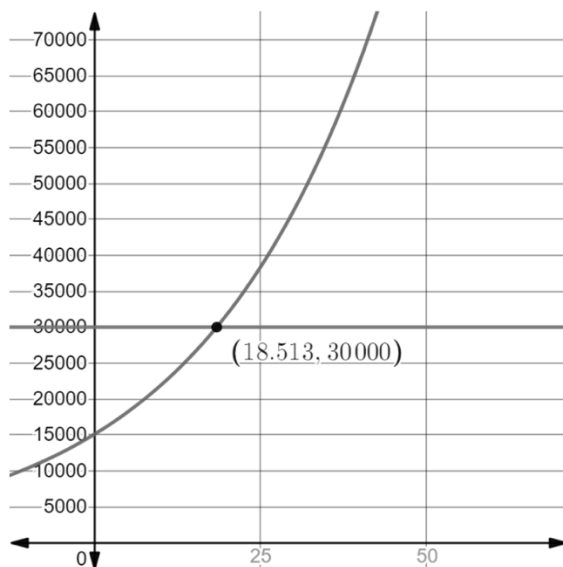
Lakita deposited \$15,000 in a savings account with a 3.75% APR compounded monthly.

1. Write an equation that models the amount of money Lakita has in the account.

2. What is the balance in the account after 10 years?

3. What is the balance in the account after 15 years?

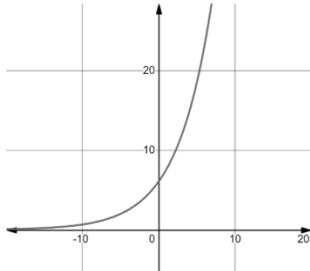
4. Use the graph to determine when the savings account will reach \$30,000.



For questions 5-10, complete the table below.

	Simple Interest	Compound Interest
Investment amount and APR	\$100,000 is invested at an interest rate of 2.25% APR	\$100,000 is invested at an interest rate of 2.25% APR compounded monthly
Write the equation.	5. $A = P(1 + rt)$	6. $A = P \left(1 + \frac{r}{n} \right)^{nt}$
Determine the amount after 5 years.	7.	8.
Determine the amount after 25 years.	9.	10.

The table shows three different functions.

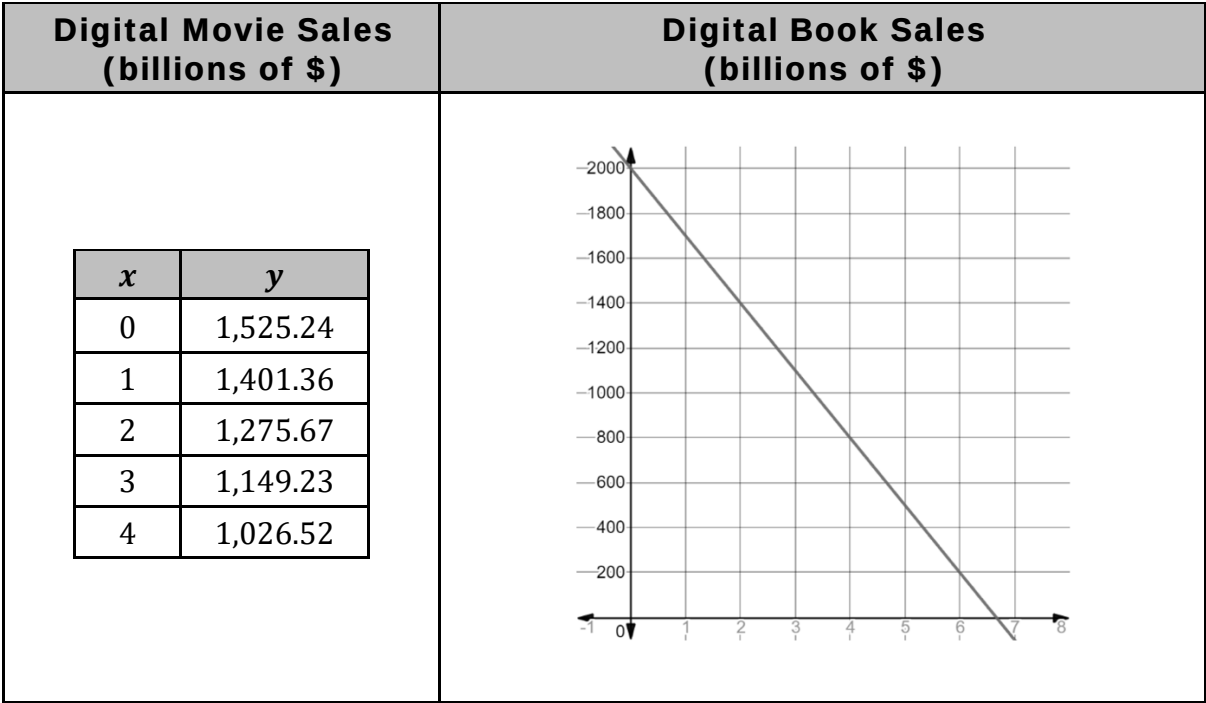
Function A	Function B	Function C												
$y = 2x + 2$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>4</td></tr><tr><td>-1</td><td>1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>4</td></tr></table>	x	y	-2	4	-1	1	0	0	1	1	2	4	
x	y													
-2	4													
-1	1													
0	0													
1	1													
2	4													

1. Determine which type of function that each represents.

Function A	Function B	Function C
☐ Exponential ☐ Linear ☐ Quadratic	☐ Exponential ☐ Linear ☐ Quadratic	☐ Exponential ☐ Linear ☐ Quadratic

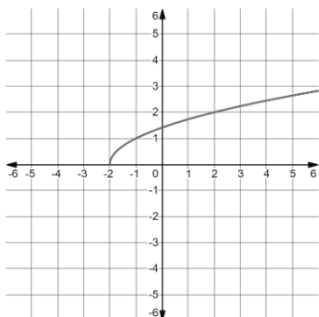
2. Which function has the largest y -intercept?
3. Compare the ranges of the functions and indicate any similarities and differences.
4. Compare the domains of the functions and indicate any similarities and differences.
5. As $x \rightarrow \infty$, which function has larger values of y ?

The table shows a table of values representing the revenue, in billions of dollars, from sales of digital movies, where $x = 0$ is 2015, and a graph representing the revenue, in billions of dollars, from sales of digital books, where $x = 0$ is 2015.

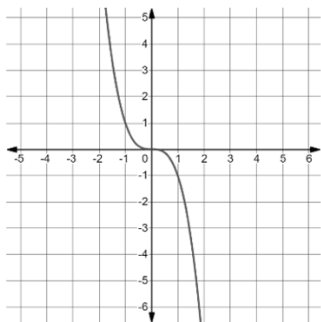


6. Approximate the x -intercept for the function that represents the sale of digital books.
7. Interpret the x -intercept of the function that represents the sale of digital books.
8. What is the domain of the function representing the sale of digital books?
9. Which function has a larger y -intercept?
10. What type of function best describes the table of values that represents digital movie sales? Explain your choice.
 - Exponential
 - Linear
 - Quadratic

Determine which family of functions each transformation is part of.

Representation of Transformation		Equation of Parent Function													
1.	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>16</td></tr><tr><td>-1</td><td>4</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>4</td></tr><tr><td>2</td><td>16</td></tr></table>	x	y	-2	16	-1	4	0	0	1	4	2	16	☐ $y = x$ ☐ $y = x^2$ ☐ $y = x^3$ ☐ $y = \sqrt{x}$	☐ $y = x$ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$
x	y														
-2	16														
-1	4														
0	0														
1	4														
2	16														
2.		☐ $y = x$ ☐ $y = x^2$ ☐ $y = x^3$ ☐ $y = \sqrt{x}$	☐ $y = x$ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$												
3.	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>5</td></tr><tr><td>-1</td><td>7</td></tr><tr><td>0</td><td>9</td></tr><tr><td>1</td><td>11</td></tr><tr><td>2</td><td>13</td></tr></table>	x	y	-2	5	-1	7	0	9	1	11	2	13	☐ $y = x$ ☐ $y = x^2$ ☐ $y = x^3$ ☐ $y = \sqrt{x}$	☐ $y = x$ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$
x	y														
-2	5														
-1	7														
0	9														
1	11														
2	13														
4.	$y = -3x^3$	☐ $y = x$ ☐ $y = x^2$ ☐ $y = x^3$ ☐ $y = \sqrt{x}$	☐ $y = x$ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$												
5.	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>9</td></tr><tr><td>-1</td><td>7</td></tr><tr><td>0</td><td>5</td></tr><tr><td>1</td><td>7</td></tr><tr><td>2</td><td>9</td></tr></table>	x	y	-2	9	-1	7	0	5	1	7	2	9	☐ $y = x$ ☐ $y = x^2$ ☐ $y = x^3$ ☐ $y = \sqrt{x}$	☐ $y = x$ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$
x	y														
-2	9														
-1	7														
0	5														
1	7														
2	9														

6.



- ☐ $y = x$
☐ $y = x^2$
☐ $y = x^3$
☐ $y = \sqrt{x}$

- ☐ $y = |x|$
☐ $y = 2^x$
☐ $y = \frac{1}{2}^x$

7.

x	-2	-1	0	1	2
y	-4	-2	-1	-0.5	-0.25

- ☐ $y = x$
☐ $y = x^2$
☐ $y = x^3$
☐ $y = \sqrt{x}$

- ☐ $y = |x|$
☐ $y = 2^x$
☐ $y = \frac{1}{2}^x$

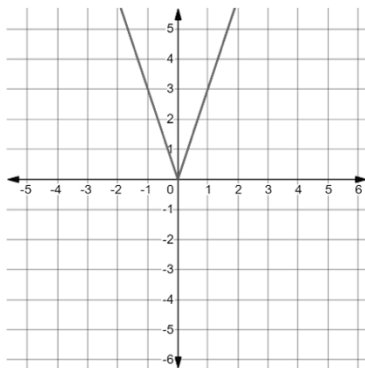
8.

x	-2	-1	0	1	2
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2

- ☐ $y = x$
☐ $y = x^2$
☐ $y = x^3$
☐ $y = \sqrt{x}$

- ☐ $y = |x|$
☐ $y = 2^x$
☐ $y = \frac{1}{2}^x$

9.



- ☐ $y = x$
☐ $y = x^2$
☐ $y = x^3$
☐ $y = \sqrt{x}$

- ☐ $y = |x|$
☐ $y = 2^x$
☐ $y = \frac{1}{2}^x$

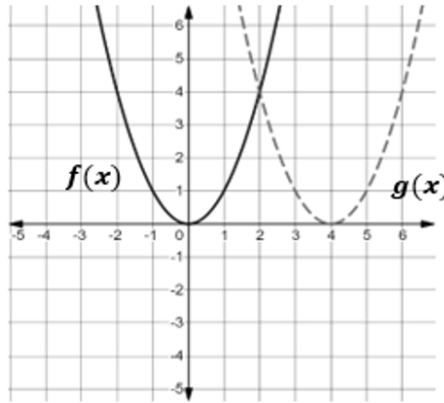
10.

$$y = -\frac{3}{4}x + 4$$

- ☐ $y = x$
☐ $y = x^2$
☐ $y = x^3$
☐ $y = \sqrt{x}$

- ☐ $y = |x|$
☐ $y = 2^x$
☐ $y = \frac{1}{2}^x$

The graphs of $y = f(x)$ and $y = g(x)$ are shown.



- Describe how to shift the graph of $f(x)$ so that it coincides with the graph of $g(x)$.
- Use the graph to complete the table of values for $f(x)$ and $g(x)$.

$f(x)$		$g(x)$	
x	y	x	y
-1		3	
0		4	
1		5	

- Explain the change in the x -values from $f(x)$ to $g(x)$.
- Complete the statement:

The graph of $g(x)$ is a
☐ horizontal
☐ vertical
 shift of $f(x)$.

- Complete the statement:

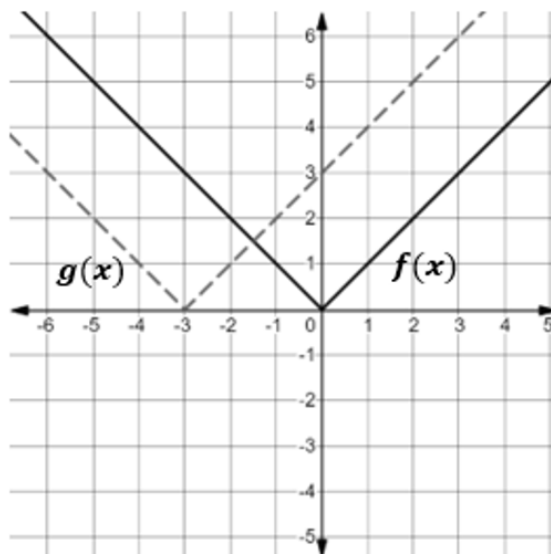
The shift is seen in the table by
☐ adding
☐ subtracting
 4 to each
☐ x -value
☐ y -value

for each corresponding
☐ x -value.
☐ y -value.

Complete the table for questions 6-8. The first row has been completed.

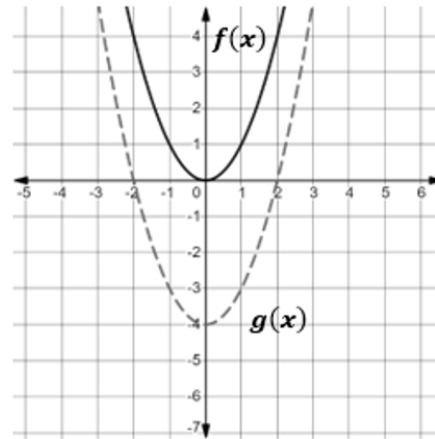
Original Equation	Original Function	Shift	Transformed Function
$y = x^2$	$f(x) = x^2$	Left 2	$f(x + 2) = (x + 2)^2$
6. $y = x $		Right 4	
7. $y = x^3$		Left 5	
8. $y = \sqrt{x}$		Right 3	

The graphs of $y = f(x)$ and $y = g(x)$ are shown.



9. How does the graph of $f(x)$ compare to the graph of $g(x)$?
10. Two function descriptions are shown. Which function description matches the transformation on the graph? Justify your choice.
- ☐ $g(x) = f(x - 3)$
 - ☐ $g(x) = f(x + 3)$

The graphs of $y = f(x)$ and $y = g(x)$ are shown.



- Describe how to shift the graph of $f(x)$ so that it coincides with the graph of $g(x)$.
- Use the graph to complete the table of values for $f(x)$ and $g(x)$.

$f(x)$		$g(x)$	
x	y	x	y
-1		-1	
0		0	
1		1	

- Explain the change in the y -values from $f(x)$ to $g(x)$.

- Complete the statement:

The graph of $g(x)$ is a
☐ horizontal
☐ vertical
 shift of $f(x)$.

- Complete the statement:

The shift is seen in the table by
☐ adding
☐ subtracting
 4 to each
☐ x -value
☐ y -value

for each corresponding
☐ x -value.
☐ y -value.

For 6-7, complete the statements to describe how the graph of $g(x) = f(x) + 7$ compares to the graph of $f(x) = |x - 1|$.

6. The graph of $g(x)$ is a

horizontal
 vertical

 shift of $f(x)$.

7. The graph of $g(x)$ is shifted _____ units

up from
 down from
 to the right of
 to the left of

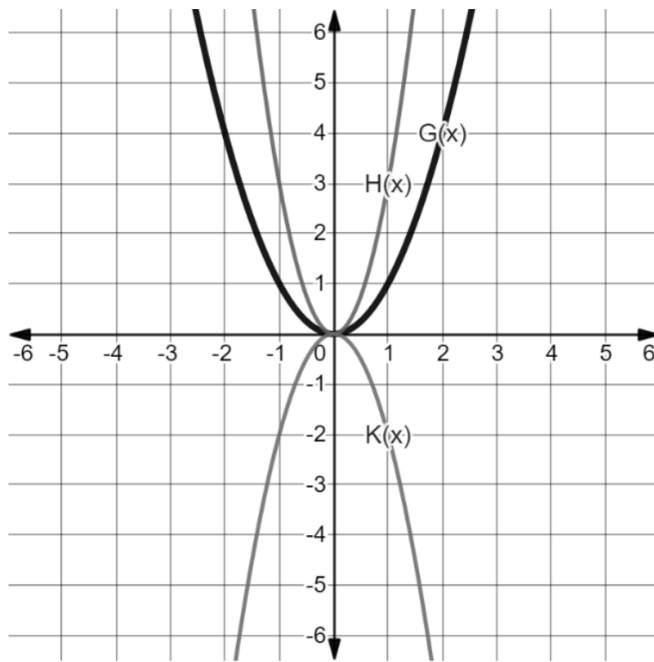
 the graph of $f(x)$.

8. Describe the effect on the x -values of $B(x) = |x + 5|$ for the transformation $C(x) = B(x) - 7$.

9. Describe the effect on the y -values of $p(x) = \frac{4}{5}x + 2$ for the transformation $r(x) = p(x) + 8$.

10. Describe how the graph of $M(x) = N(x + 10)$ compares to the graph of $N(x) = x^2 + 3$.

The functions $G(x)$, $H(x)$, and $K(x)$ are shown on the graph, where $G(x) = x^2$, $H(x) = 3x^2$, and $K(x) = -2x^2$. The table of values for each function are also shown.



$y = G(x)$		$y = H(x)$		$y = K(x)$	
x	y	x	y	x	y
-3	9	-3	27	-3	-18
-2	4	-2	12	-2	-8
-1	1	-1	3	-1	-2
0	0	0	0	0	0
1	1	1	3	1	-2
2	4	2	12	2	-8
3	9	3	27	3	-18

Complete the statements below.

- The coefficient of $G(x) = x^2$ is _____.
- The coefficient of $H(x) = 3x^2$ is _____.
- The y -values of $H(x)$ can be found by multiplying the y -values of $G(x)$ by _____.
- The coefficient of $K(x) = -2x^2$ is _____.
- The y -values of $K(x)$ can be found by multiplying the y -values of $G(x)$ by _____.

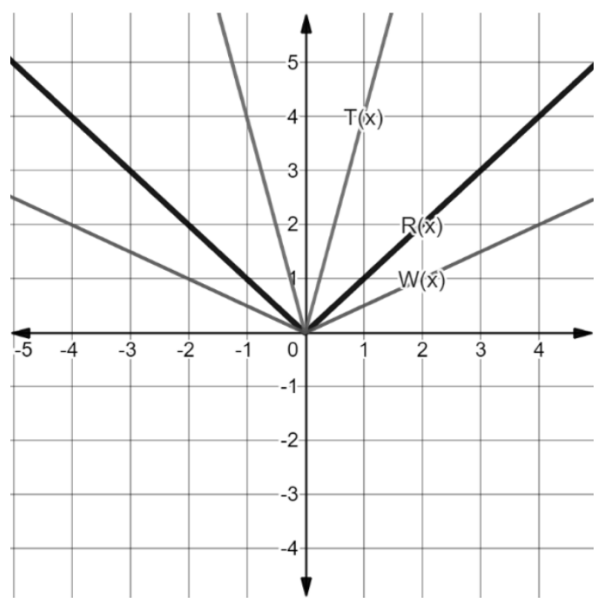
The table of values of the function $y = A(x)$ is shown.

x	-6	-4	-2	0	2	4	6
y	4	7	10	13	16	19	22

Complete the sentence to describe the effect on the x -values for the function $y = A(3x)$.

6. The x -values for $A(x)$ will be
☐ multiplied
☐ divided
 by 3.

The functions $R(x)$, $T(x)$, and $W(x)$ are shown on the graph, where $R(x) = |x|$, $T(x) = |4x|$, and $W(x) = |0.5x|$. The table of values for each function are also shown.



$y = R(x)$		$y = T(x)$		$y = W(x)$	
x	y	x	y	x	y
-2	2	$-\frac{1}{2}$	2	-4	2
-1	1	$-\frac{1}{4}$	1	-2	1
0	0	0	0	0	0
1	1	$\frac{1}{4}$	1	2	1
2	2	$\frac{1}{2}$	2	4	2

Complete the statements below.

7.

The x -values of $T(x)$ can be found by _____.

☐ multiplying

☐ dividing

the x -values of $R(x)$ by _____.
8.

The x -values of $W(x)$ can be found by _____.

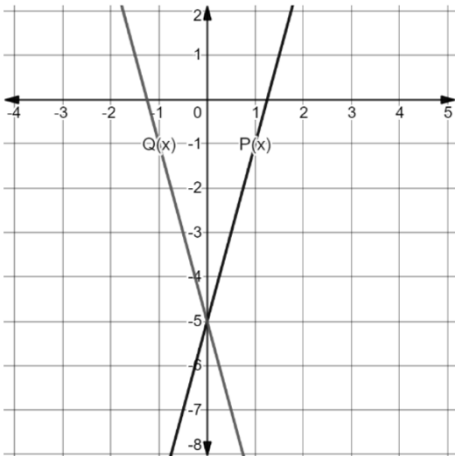
☐ multiplying

☐ dividing

the x -values of $R(x)$ by _____.

The functions $P(x)$ and $Q(x)$ are shown on the graph where $Q(x) = P(-x)$. The table of values for each function are also shown.

$y = P(x)$		$y = Q(x)$	
x	y	x	y
-2	-13	-2	3
-1	-9	-1	-1
0	-5	0	-5
1	-1	1	-9
2	3	2	-13



9.
- What is the effect on the table of values when the x in $P(x)$ was replaced with $-x$ to create $Q(x)$?

10.

What is the effect on the graph when the x in $P(x)$ was replaced with $-x$ to create $Q(x)$?

Suni is biking 20 miles. To fuel her bike ride, she will need to consume 738 calories. She plans to bring sports drink and granola bars. Sports drink bottles are 158 calories and granola bars are 132 calories.

1. $158s + 132g$ is a(n)

- ☐ function.
- ☐ equation.
- ☐ expression.

2. The values 158 and 132 are

- ☐ coefficients.
- ☐ terms.
- ☐ constants.

3. The monomials $158s$ and $132g$ are

- ☐ coefficients.
- ☐ terms.
- ☐ constants.

4. $158s + 132g = 738$ is a(n)

- ☐ function.
- ☐ equation.
- ☐ expression.

5. Determine the amount of granola bars Suni will need if she has three bottles of sports drink.

A rancher has 3000 feet of fencing to enclose his property, which borders a lake. The diagram shows the width, w , of the property and the lake that his property borders. The fencing is not used along the lake.

Explain what each quantity represents in this context.

6. $3000 - 2w$

7. $w(3000 - 2w)$



Luis started a retirement savings fund when he began his first job in 2015. The amount of money in the retirement savings fund since 2015 can be represented by the equation $y = 15,000(1 + 0.025)^x$, where y is the amount in the retirement savings fund and x is the number of years since 2015.

8. The equation $y = 15,000(1 + 0.025)^x$ represents

- ☐ exponential growth.
- ☐ exponential decay.

9. The rate of ☐ decay ☐ growth is ____%.

10. What does the coefficient 15,000 represent?

Algebra 1

Unit 13

Lesson 9 Homework

Name_____

Date_____

Period_____

To make a 15% acid solution, a chemist mixes 3 parts of 4% acid solution with a 20% acid solution. The expression $3(0.04x) + 0.20y$ represents the mixture, where x is the number of ounces of 4% solution and y is the number of ounces of the 20% solution.

Explain what each quantity means.

1. $(0.04x)$

2. $3(0.04x)$

3. $0.20y$

Since 2015, the number of independently contracted food delivery drivers can be modeled by the equation $y = 1,250,000(1.073)^x$.

4. What does the x variable represent?

5. What does the y variable represent?

6. What does the quantity 1,250,000 represent?

7. What is the constant percent rate of change?

The equation $y = -2(x - 4)^2 + 45$ models the height, y , in feet, of a ball, x seconds after it is thrown into the air.

Explain what each quantity represents.

8. $x - 4$

9. 45

Jovi is buying cereal for \$3.25 per box. He has a coupon for two free cereals. He pays \$4.75 for a gallon of milk. The expression $3.25(x - 2) + 4.75$ can be used to represent the overall cost.

10. What does the quantity $3.25(x - 2)$ represent?

Algebra 1

Unit 13

Lesson 10 Homework

Name _____

Date _____

Period _____

Given a written description, identify the function as linear, quadratic, or exponential. Explain your choice.

1. An investment grows at a constant percent rate of 2.25% APR.
2. The Occupational Safety and Health Administration (OSHA) allows for stairs to have a rise of 9 inches and a run of 10 inches.
3. A ball is thrown from a height of 5 feet. Four seconds later, it reaches a maximum height of 25 feet.
4. A rare species of fish experiences a population decline at a constant percent rate of 4.5%.

Given a table of values, identify the function as linear, quadratic, or exponential. Explain your choice.

5.

x	y
0	120
1	111
2	102
3	93
4	84

6.

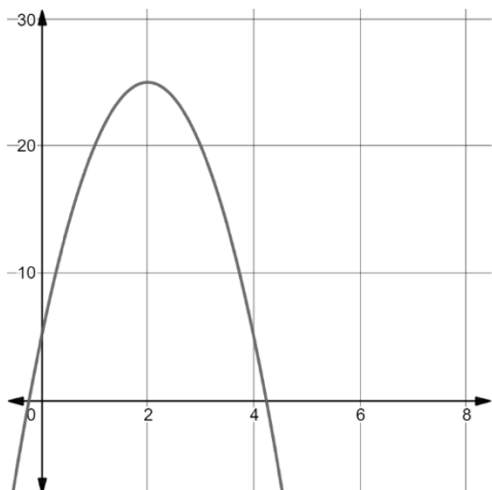
x	y
0	165
1	179
2	195
3	211
4	230

7.

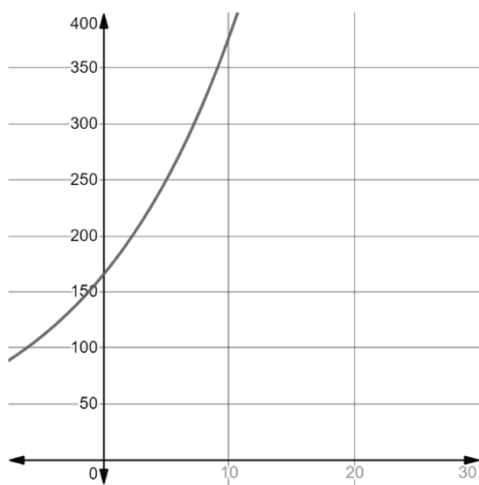
x	y
1	20
2	35
3	40
4	35
5	20

Given a graph, identify the function as linear, quadratic, or exponential. Explain your choice.

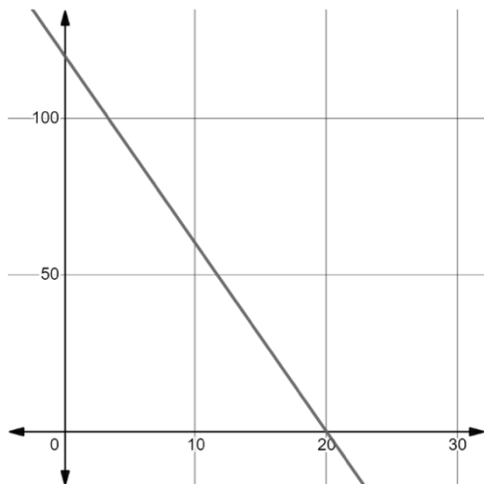
8.



9.



10.



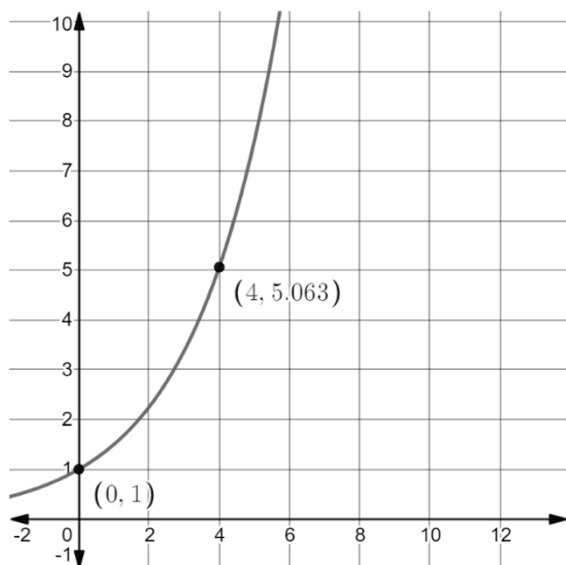
The table shows the number of tons of blueberries that were imported to the US from 2005 to 2013.

Year	Blueberries Imported (tons)
2005	26,319
2006	32,602
2007	35,125
2008	52,111
2009	60,450
2010	76,743
2011	87,336
2012	96,367
2013	103,502

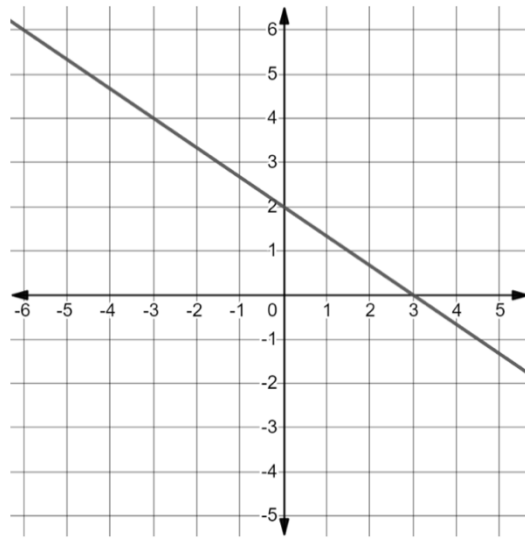
1. Determine the average rate of change from 2005 to 2013.
2. Interpret the average rate of change for the interval 2005 to 2013 in terms of the context.
3. Determine the average rate of change from 2007 to 2010.
4. Interpret the average rate of change for the interval 2007 to 2010 in terms of the context.
5. How does the average rate of change for the interval 2007 to 2010 compare to the average rate of change for the interval 2005 to 2013?

The function $M(x) = 5x^2 - 160x + 6179$ models the number of miles, in millions, U.S. passengers traveled on a train for the period of 1970 to 2000, where x is the number of years since 1970.

6. What is the average rate of change, in miles traveled by U.S. passengers per year, for the period 1970 to 1975?
7. Interpret the average rate of change for the interval 1970 to 1975 in terms of the context.
8. What is the average rate of change, in miles traveled by U.S. passengers per year, for the period 1975 to 1980?
9. Interpret the average rate of change for the interval 1975 to 1980 in terms of the context.
10. The graph of a function is shown. Points $(0, 1)$ and $(4, 5.063)$ are on the function. Calculate the average rate of change for the line that passes through the points $(0, 1)$ and $(4, 5.063)$.



The graph of the function $P(x)$ is shown.

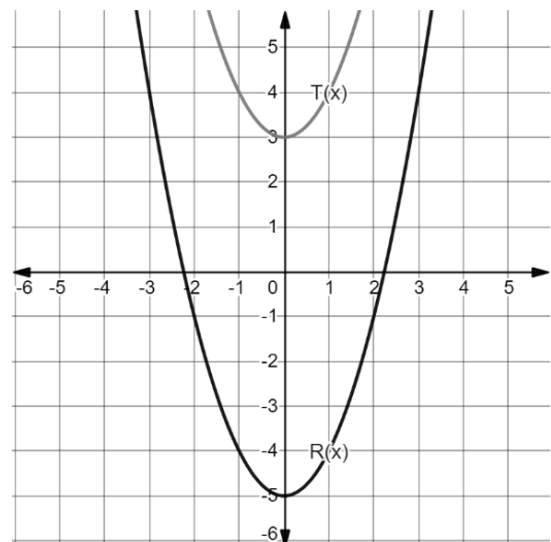


1. Write the function, $P(x)$, that represents the line.
2. Draw a line that is parallel to $P(x)$ with a y -intercept of -2 . Label this line $Q(x)$.
3. Describe how $P(x)$ can be shifted to coincide with $Q(x)$.
4. Write the function $Q(x)$ in terms of $P(x)$.

5. The graphs of $R(x)$ and $T(x)$ are shown on the graph.

Write the transformation by completing the equation.

$$T(x) = R(x) + \underline{\hspace{2cm}}$$

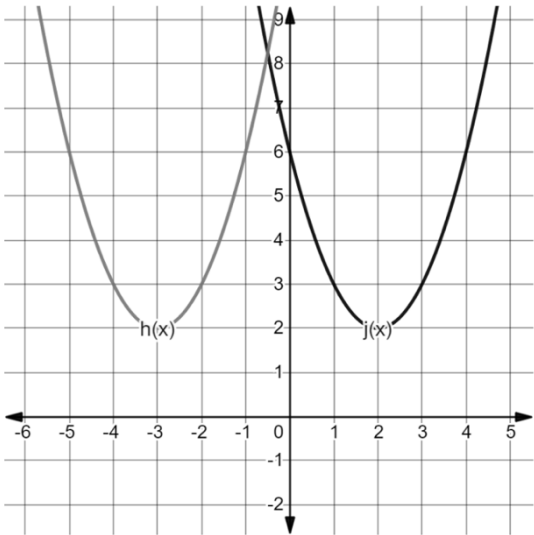


The functions $h(x)$ and $j(x)$ are shown on the graph.

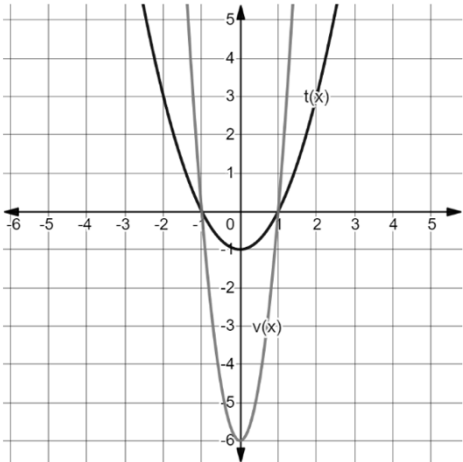
6. Write $h(x)$ in vertex form.

7. Write $j(x)$ in vertex form.

8. Describe how $h(x)$ was transformed to create $j(x)$.



9. The functions $t(x)$ and $v(x)$ are shown on the graph, where $v(x)$ is a transformation of $t(x)$. What is the value of k such that $v(x) = kt(x)$?



10. The table of values for $w(x)$ and $z(x)$ are shown, where $z(x)$ is a transformation of $w(x)$. Write a description of the transformation.

$y = w(x)$		$y = z(x)$	
x	y	x	y
-2	8	-2	13
-1	2	-1	7
0	0	0	5
1	2	1	7
2	8	2	13

Algebra 1

Unit 13

Lesson H2 Homework

Name _____

Date _____

Period _____

Use the functions $k(x) = 3x^2 + 4x - 5$ and $p(x) = 6 - 2x$ to determine each new function. Write your answer in standard form.

1. $(k + p)(x)$

2. $(k - p)(x)$

3. $(p - k)(x)$

4. $(k \cdot p)(x)$

5. $\left(\frac{p}{p}\right)x$

6. Write the restriction on the domain for $\left(\frac{p}{p}\right)x$.

Juan and Marina are renting a car for a cross country road trip. The car costs \$155 per day. The rental includes 100 miles for free and then charges \$0.25 per mile. The function $C(m) = 0.25(m - 100) + 155$ models the cost of the car, C , in terms of miles traveled, m . They determine gas will cost \$1.15 for each mile they travel. The company charges a \$40 fee for the first tank of gas. The function $G(m) = 1.15m + 40$ models the cost of gasoline for their vacation.

7. What is a reasonable domain for $C(m)$?

8. What is a reasonable domain for $G(m)$?

9. Determine the function that gives the total cost, $T(m)$, in dollars, based on the miles traveled.

10. What is a reasonable domain for $T(m)$?